

CAM-IMX335-5MP User Manual



Date	Version
2025/12/05	V1.0
2026/4/16	V1.1

1. Product Description

1.1 Overview

The CAM-IMX335-5MP is a high-performance 5MP camera module developed exclusively for the Raspberry Pi platform (Raspberry Pi 5 / 4B / 3B+ / 3B / Zero 2W / Zero W / CM4 / CM5).

It utilizes the authentic Sony STARVIS IMX335 5.14-megapixel back-illuminated sensor and ships with a factory-optimized tuning file, paired with a high-quality lens offering a wide field of view (H 86.7° / V 48.8° / D 99.5°, F/1.8, Focal Length 4.9mm).

Requires recompiling the libcamera library and rpicas applications to enable IPA helper support for the IMX335 sensor. We provide compilation scripts for libcamera and rpicas on latest raspberry pi Trixie, as well as pre-compiled system IMG files.

Supported and fully validated mode is **2592 × 1944 (True 5MP) 30 fps**. Ideal for 5MP AI vision, high-resolution surveillance, industrial AOI, machine vision, scientific imaging and long-term time-lapse projects requiring broad coverage with the specified FOV.

1.2 Key Features

- Genuine Sony STARVIS IMX335 sensor – real hardware 5MP
- Officially supported mode: **2592 × 1944 @ 30 fps**
- Starlight-level full-color imaging down to 0.1 Lux
- Wide lens field of view: H 86.7° / V 48.8° / D 99.5°
- Built-in IR filter
- Native support through official libcamera / rpicas-apps (dtoverlay=imx335 + tuning file required); requires recompiling the libcamera library and rpicas applications to enable IPA helper support for the IMX335 sensor. We provide compilation scripts for libcamera and rpicas, as well as pre-compiled system IMG files.
- Full 2-lane CSI-2 bandwidth utilization on Raspberry Pi 5
- DOL-HDR up to 110 dB
- Low power consumption

2. Technical Specifications

2.1 Sensor

Item	Specification
Sensor Model	Sony IMX335 (STARVIS series)
Sensor Type	1/2.8" Back-illuminated stacked CMOS
Total Pixels	2616 (H) × 1964 (V) = 5.14 Megapixels
Raspberry Pi Usable Area	2592 × 1944
Pixel Size	2.0 μm × 2.0 μm
Shutter Type	Electronic rolling shutter
Output Format	RAW10 (default) / RAW12
Max SNR	37 dB
Dynamic Range	71 dB (standard), up to 110 dB (DOL-HDR)
Sensitivity	0.1 Lux @ F1.8 (full-color starlight)

2.2 Lens Optics

Chapter 3.3

2.3 Electrical

Item	Specification
Interface	MIPI CSI-2
Power Supply	3.3 V (directly from Raspberry Pi CSI port)
I2C Control	Yes

2.4 Mechanical

Item	Specification
Module Dimensions	32 × 32 mm
Total Height	≤ 20 mm (final height depends on selected)

InnoMaker	CAM-IMX335-5MP
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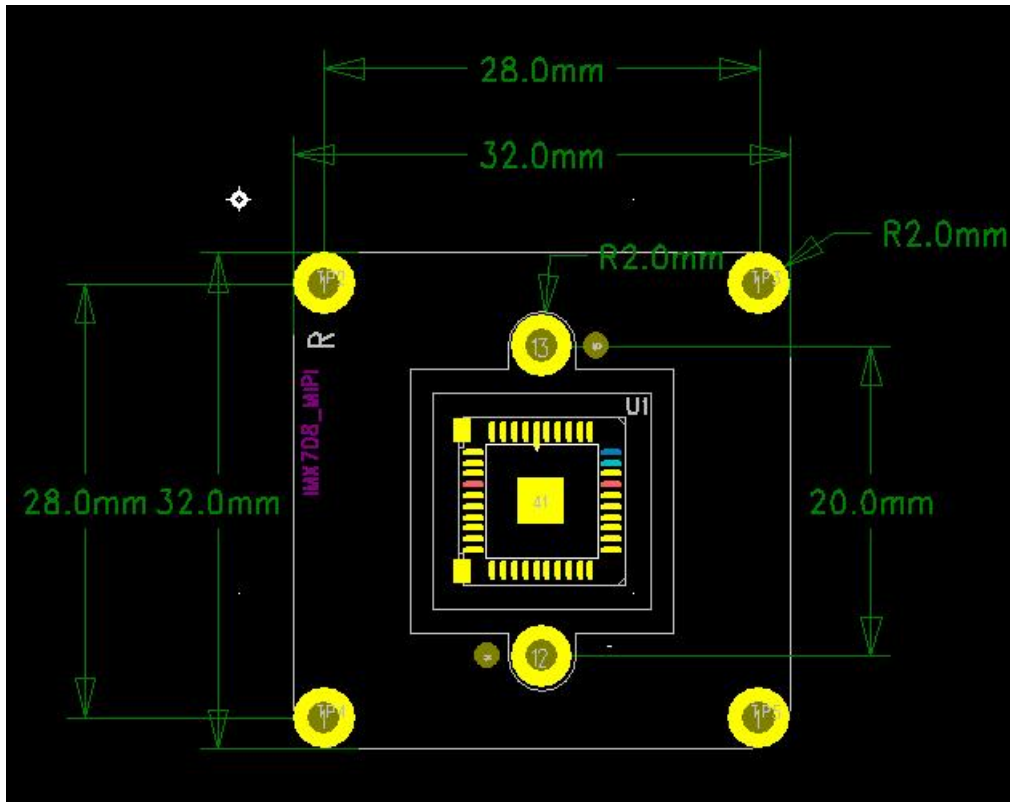
	lens)
Weight	≈12–15 g (including flex cable)
Flex Cable	150 mm 15-pin 1.0 mm + 150 mm 22-pin 0.5 mm
Mounting Holes	4 × M2, compatible with all official cases & mounts

2.5 Environmental

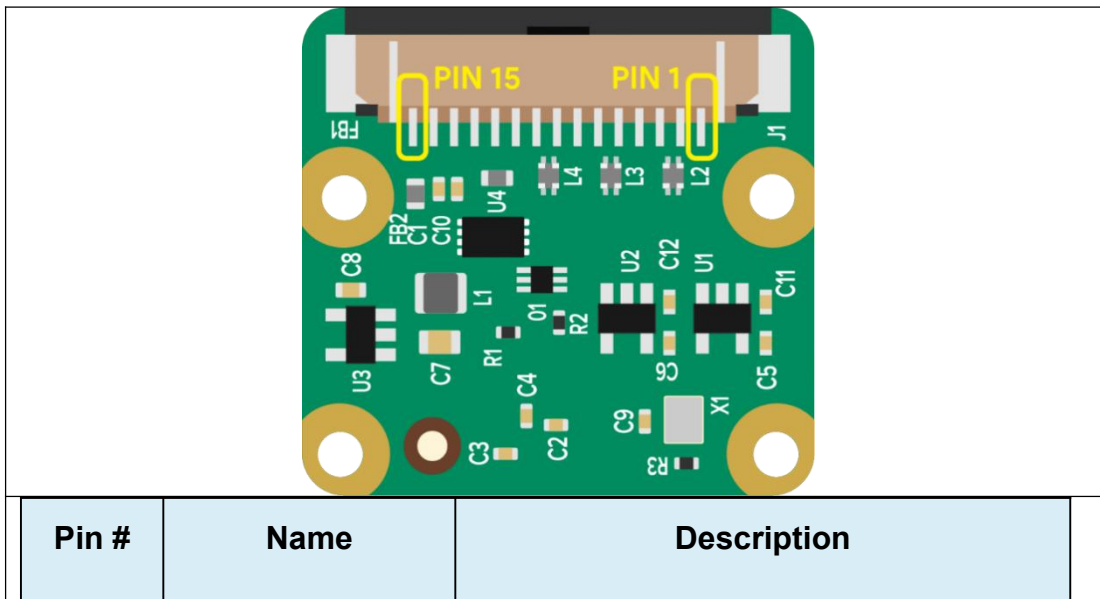
Item	Specification
Operating Temperature	–20 °C to +70 °C
Storage Temperature	–30 °C to +85 °C
Humidity	20 %–85 % RH (non-condensing)

3 Hardware

3.1 Camera Size



3.2 Camera Module Pins Out



1	GND	Ground
2	CAM_D0_N	MIPI Data Lane 0 Negative
3	CAM_D0_P	MIPI Data Lane 0 Positive
4	GND	Ground
5	CAM_D1_N	MIPI Data Lane 1 Negative
6	CAM_D1_P	MIPI Data Lane 1 Positive
7	GND	Ground
8	CAM_CK_N	MIPI Clock Lane Negative
9	CAM_CK_P	MIPI Clock Lane Positive
10	GND	Ground
11	CAM_IO0	Power Enable
12	CAM_IO1	LED Indicator
13	CAM_SCL	I2C SCL
14	CAM_SDA	I2C SDA
15	CAM_3V3	3.3V Power Input

3.3 Lens Information

5 主要参数 (Main Items)

5.1 光学参数规格 (Optical Specifications)

M12 4mm 1/1.8

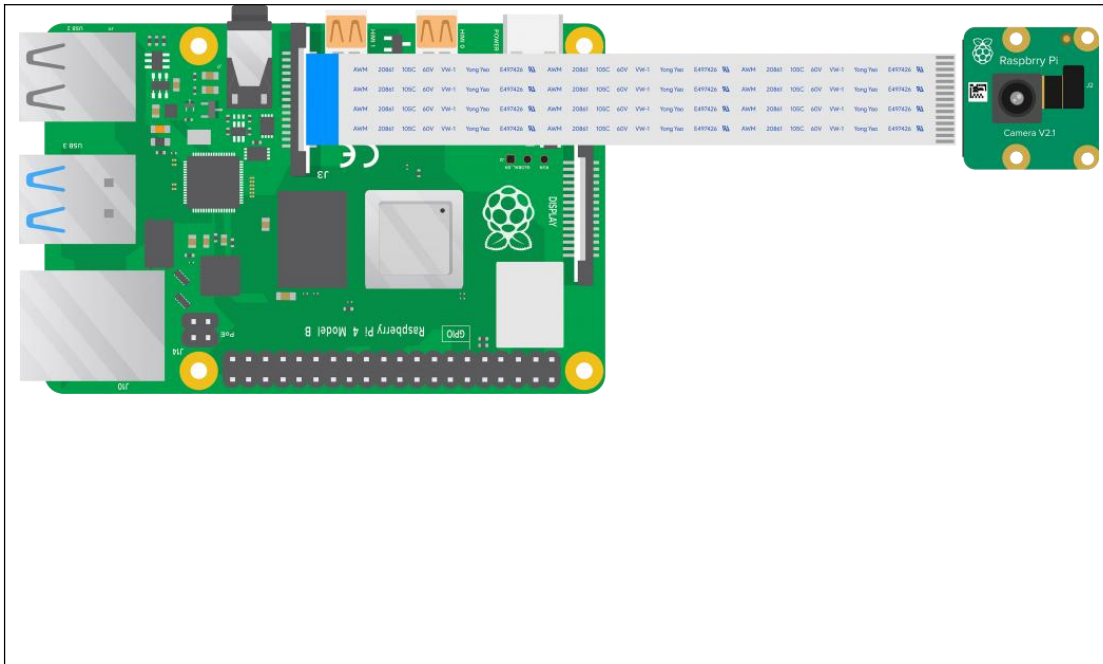
NO.	项目 (Items)		具体规格 (Specification)		
1	F No.		1.65±10%		
2	焦距(Focal-Length)		4.9±5%		
3	光学后焦 (Optical Back Focal Length)		6.41±0.2 (in air)		
4	机械后焦 (Mechanical Back Focal Length)		6.17±0.2 (in air)		
5	镜头总长(TTL)		22.28±0.2 (in air)		
6	像面大小 (Image circle)		Φ9.2 (MAX)		
7	镜片构成 (Lens structure)		4G4P		
8	接口 (Mount)		M12*P0.5		
9	镜头与底座螺纹配合扭力 (Clamp Force)		60-600gf.cm		
10	视场角(FOV)	sensor型号	H(水平)	V(垂直)	D(对角)
		1/1.8"(16:9)	86.7°	48.8°	99.5°
		1/2.5"(16:9)	70.2°	39.5°	80.6°
		1/2.7"(16:9)	65°	36.6°	74.6°
11	光学畸变 (Optical Distortion)		1/1.8"	1/2.5"	1/2.7"
			-25.6%	-17.6%	-14.8%
12	相对亮度 (Relative Illumination)		1/1.8"	1/2.5"	1/2.7"
			40%	58%	60%
13	最大主光线夹角 (CRA)		1/1.8"	1/2.5"	1/2.7"
			16.2°	12.3°	11.9°
14	近摄距 (M.O.D)		0.6m		
15	解像标准 (Resolution)		分辨率 (Resolution): 3840*2160 (8MP)		
16	重量 (Weight)		/		
17	操作方法 (Operation)		聚焦 (Focus)	手动 (Manual)	
			光圈 (Iris)	固定 (Fixed)	
18	环保&安全 (HSF&Safety)		RoHS		

Apply for IMX335 Sensor Module

Sensor	Sensor size / typical mode	Focal Length	Aperture	Horizontal FOV	Vertical FOV	Diagonal FOV
IMX335	1/2.8" (≈ 1/1.8" mode, 16:9)	4.9 mm (±5%)	f/1.8	86.7°	48.8°	99.5°

IMX415	1/2.8" ($\approx$ 1/2.5" mode, 16:9)	4.9 mm ($\pm 5\%$)	f/1.8	70.2°	39.5°	80.6°
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3.4 Connection



4 Build libcamera/rpicam

4.1 Method 1(Recommend)

Build from local source

```

sudo apt-get update

sudo apt-get dist-upgrade

sudo git clone https://github.com/INNO-MAKER/CAM-IMX335-5MP.git

cd CAM-IMX335-5MP

sudo chmod -R a+rw *

sudo tar -zxvf build_offline_complete.tar.gz

cd build_offline_complete

```

```
sudo ./build offline complete trixie.sh #install dependent
```

After Finish, Change config.txt accoring to Chapter5.

4.2 Method 2 (Recommend)

Directly install precompiler files.

```
sudo git clone https://github.com/INNO-MAKER/CAM-IMX335-5MP.git
cd CAM-IMX335-5MP
sudo chmod -R a+rwX *
sudo ./replace_ipa.sh
```

4.3 Method 3 (Not Recomend)

Build from download source

4.3.1 Download scripts

```
sudo apt-get update
sudo apt-get dist-upgrade
sudo git clone https://github.com/INNO-MAKER/CAM-IMX335-5MP.git
cd CAM-IMX335-5MP
sudo chmod -R a+rwX *
sudo tar -zxvf build_online_complete.tar.gz
cd build_online_complete
```

4.3.2 Install dependent

```
sudo ./install_dependent.sh #install dependent
```

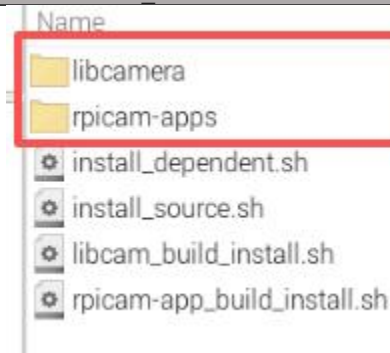
```

0 upgraded, 3 newly installed, 0 to remove and 34 not upgraded.
Need to get 2,955 kB of archives.
After this operation, 8,900 kB of additional disk space will be used.
Get:1 http://archive.raspberrypi.com/debian bookworm/main arm64 libpostproc-dev arm64 8:5.1.7-0+deb12u1+rpt1 [87.1 kB]
Get:2 http://archive.raspberrypi.com/debian bookworm/main arm64 libavfilter-dev arm64 8:5.1.7-0+deb12u1+rpt1 [2,693 kB]
Get:3 http://archive.raspberrypi.com/debian bookworm/main arm64 libavdevice-dev arm64 8:5.1.7-0+deb12u1+rpt1 [175 kB]
Fetched 2,955 kB in 4s (714 kB/s)
Selecting previously unselected package libpostproc-dev:arm64.
(Reading database ... 162531 files and directories currently installed.)
Preparing to unpack .../libpostproc-dev_8%3a5.1.7-0+deb12u1+rpt1_arm64.deb ...
Unpacking libpostproc-dev:arm64 (8:5.1.7-0+deb12u1+rpt1) ...
Selecting previously unselected package libavfilter-dev:arm64.
Preparing to unpack .../libavfilter-dev_8%3a5.1.7-0+deb12u1+rpt1_arm64.deb ...
Unpacking libavfilter-dev:arm64 (8:5.1.7-0+deb12u1+rpt1) ...
Selecting previously unselected package libavdevice-dev:arm64.
Preparing to unpack .../libavdevice-dev_8%3a5.1.7-0+deb12u1+rpt1_arm64.deb ...
Unpacking libavdevice-dev:arm64 (8:5.1.7-0+deb12u1+rpt1) ...
Setting up libpostproc-dev:arm64 (8:5.1.7-0+deb12u1+rpt1) ...
Setting up libavfilter-dev:arm64 (8:5.1.7-0+deb12u1+rpt1) ...
Setting up libavdevice-dev:arm64 (8:5.1.7-0+deb12u1+rpt1) ...

```

4.3.3 Install Source of libcamera and rpicam

```
sudo ./install_source.sh
```



After downloading successfully, there will be 2 new folder displays.

Copy the scripts to the folder accordingly.

```

sudo cp libcam_build_install.sh libcamera/           # copy scripts
sudo cp rpicam-app_build_install.sh rpicam-apps/     # copy scripts
sudo cp meson.build libcamera/src/ipa/rpi/cam_helper/ # copy meson_build
sudo cp cam_helper_imx335.cpp libcamera/src/ipa/rpi/cam_helper/ # copy imx335.cpp

```

4.3.4 Compiler libcamera libraries

```
cd libcamera
```

```
sudo ./libcam_build_install.sh
```

```

Installing /home/pi/CAM-IMX335-5MP/libcamera/src/ipa/rpi/pisp/data/vd86g3.json to /usr/local/share/libcamera/ipa/rpi/pisp
Installing /home/pi/CAM-IMX335-5MP/libcamera/src/ipa/rpi/pisp/data/vd86g3_mono.json to /usr/local/share/libcamera/ipa/rpi/pisp
Installing /home/pi/CAM-IMX335-5MP/libcamera/src/py/libcamera/_init_.py to /usr/local/lib/python3/dist-packages/libcamera
Installing /home/pi/CAM-IMX335-5MP/libcamera/build/src/v4l2/libcamerapy to /usr/local/bin
Installing symlink pointing to libcamera-base.so.0.6.0 to /usr/local/lib/aarch64-linux-gnu/libcamera-base.so.0.6
Installing symlink pointing to libcamera-base.so.0.6 to /usr/local/lib/aarch64-linux-gnu/libcamera-base.so
Installing symlink pointing to libcamera.so.0.6.0 to /usr/local/lib/aarch64-linux-gnu/libcamera.so.0.6
Installing symlink pointing to libcamera.so.0.6 to /usr/local/lib/aarch64-linux-gnu/libcamera.so
Installing custom install script /home/pi/CAM-IMX335-5MP/libcamera/src/ipa/sign-install.sh /home/pi/CAM-IMX335-5MP/libcamera/build/src/ipa-pri-key.pem lib/aarch64-linux-gnu/libcamera/ipa/ipa_v4l2.so lib/aarch64-linux-gnu/libcamera/
Generating IPA module signatures
Running custom install script /usr/bin/python3 /home/pi/CAM-IMX335-5MP/libcamera/build/meson-private/pycompile.py python-3.12-installed.json 0
Compiling /usr/local/lib/python3/dist-packages/libcamera/_init_.py...

```

4.3.5 Compiler rpicas-app

```
cd rpicas-apps/
```

```
sudo ./rpicas-app build install.sh
```

```
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/post_processing_stages/segmentation.hpp to /usr/local/include/rpicas-apps/post_processing_stages
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/post_processing_stages/tf_stage.hpp to /usr/local/include/rpicas-apps/post_processing_stages
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/preview/preview.hpp to /usr/local/include/rpicas-apps/preview
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/assets/hdr.json to /usr/local/share/rpi-camera-assets
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/assets/motion_detect.json to /usr/local/share/rpi-camera-assets
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/assets/negate.json to /usr/local/share/rpi-camera-assets
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/assets/acoustic_focus.json to /usr/local/share/rpi-camera-assets
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/utis/camera_bug_report to /usr/local/bin
Installing /home/pi/CAM-IMX335-5MP/rpicas-apps/build/meson-private/rpicas_app.pc to /usr/local/lib/aarch64-linux-gnu/pkgconfig
Installing symlink pointing to librpicas_app.so.1.10.1 to /usr/local/lib/aarch64-linux-gnu/librpicas_app.so.1
Installing symlink pointing to librpicas_app.so.1 to /usr/local/lib/aarch64-linux-gnu/librpicas_app.so
```

5. Quick Start Guide

5.1 Edit config file

```
sudo nano /boot/firmware/config.txt # Pi 5
# or
sudo nano /boot/config.txt # Pi 4/3/Zero
```

5.2 Add these two lines at the bottom:

```
camera_auto_detect=0
dtoverlay=imx335,cam0 # for pi5 cam0 is a must.
```

Save and reboot

```
sudo reboot
```

5.3 Test commands:

```
rpicas-hello -t 0 # Live preview

rpicas-hello -t 0 --camera 0 # Point channel if Dual camera

rpicas-hello -t 0 --camera 1 # Point channel if Dual camera

rpicas-hello --list-cameras # list cameras

rpicas-hello --list-cameras -v # list camera rw value

rpicas-still -o 4k.jpg --width 2592 --height 1944 # Capture photo

rpicas-vid -t 20000 --width 2592 --height 1944 -o xxx.h264 # Record 20 s 5mp video
```

6 Official Documentation

- [Camera Description on Raspberry Pi](#)
- [Camera Software On Raspberry Pi](#)